Device-Side Invalidation Ordering: Testing IOMMU Translation Caches at RTL

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Abstract—When an operating system revokes a device's access to a page it unmaps the page-table entry, invalidates the IOMMU's translation cache, and waits for the invalidation to complete before reusing the page. The interval between the unmap and the completion of that wait is a containment window: if the IOMMU answers a device transaction from a stale cached translation after the wait has returned, the device writes memory that has already been handed to someone else. Memory-ordering verification has studied the analogous question for CPU MMUs, and RTL-level tools exist for processor consistency models, but we are not aware of work that tests this obligation on the device side, against IOMMU hardware. We define six ordering properties for translation-cache invalidation—staleness permitted before invalidation, effectiveness, scope precision, scope containment, completion ordering, and completion ordering under concurrent traffic—express them as executable tests over an IOMMU's command interface, and run them against an open RISC-V IOMMU implementation, including a case with twenty-four device transactions in flight across the invalidation. The implementation satisfies all of them. To show the tests discriminate rather than merely pass, we inject two invalidation defects into the RTL; both are detected. We also report a structural observation: the implementation's fence does not wait for preceding commands, and its ordering guarantee therefore holds because every command it implements completes immediately. That is a property of the current feature set, not of the design, and it would not survive the addition of a long-latency command such as an ATS invalidation.

Index Terms—IOMMU, translation cache, invalidation, memory ordering, RISC-V, RTL verification, DMA.

I. Introduction

THE correctness of DMA confinement rests on an ordering argument that software makes and hardware must honour. To revoke a device's access to a page, an operating system clears the page-table entry, issues an invalidation to the IOMMU, waits for that invalidation to be reported complete, and only then reuses the page. Every step of that sequence is an ordering obligation on the IOMMU: after the completion is observable, no subsequent device transaction may be answered from a translation that the invalidation was supposed to remove.

If the obligation is not met, the failure is not a performance anomaly. The device holds a translation to a physical page that now belongs to another domain, and its next write lands there. It is a use-after-free that crosses a protection boundary, and it is invisible to the software that performed the unmap, because from software's point of view the invalidation returned.

It is worth being concrete about who is exposed. The scenario is not exotic: a network interface or storage controller is given a DMA buffer, the driver finishes with it and unmaps it, the allocator hands the page to something else, and the device—still holding a cached translation, because the IOMMU's invalidation did not take effect when it said it had—completes an outstanding write into it. The device need not be malicious for this to corrupt memory, and if it is malicious, or if its firmware has been compromised, the window is a capability rather than an accident: peripheral-originated attacks against IOMMU protection are a studied threat [6], and they are studied on the assumption that the hardware honours the invalidations the operating system issues.

The analogous question on the CPU side has been studied carefully. Memory consistency models describe the ordering of loads and stores; translation-aware extensions describe what happens when the mappings themselves change [1], [2]; and tools exist to check that RTL satisfies the microarchitectural axioms those models assume [3], [4]. All of this concerns a processor's MMU, its TLBs and the software that maintains them. The device side—an IOMMU's translation caches, the command queue that invalidates them, and the fence that reports completion—has not, to our knowledge, been given the same treatment against hardware.

This article is a focused study of that gap. It does not propose a formal model. It defines a small family of ordering properties that are meaningful for IOMMU translation caches, makes them executable against real RTL, reports the result on an open implementation, and demonstrates that the tests have the power to catch violations by injecting them.

A. Contributions

1) Six ordering properties for IOMMU translation-cache invalidation, stated operationally so that each can be decided by observing an IOMMU's completion port.
2) An executable realisation of them against an open RISC-V IOMMU, including an invalidation performed with device traffic in flight throughout.
3) A fault-injection check that the tests discriminate: two injected invalidation defects are both detected.
4) A structural observation about the implementation's fence, whose ordering guarantee is currently incidental rather than architected.

II. Background

A. What the hardware provides

A RISC-V IOMMU caches completed translations in an IOTLB and device contexts in directory caches. Software manages them through a command queue. An IOTINVAL.VMA command invalidates first-stage translations; it may be unscoped, or scoped by address, by address-space identifier, or by virtual machine identifier. An IOFENCE.C command is the completion primitive: when its address-valid bit is set, the IOMMU writes a software-supplied data word to a software-supplied address once preceding commands have taken effect. The appearance of that word in memory is what software waits on.

B. What software expects

The unmap path in an operating system is, in order: clear the page-table entry; enqueue an invalidation covering the affected address; enqueue a fence; wait for the fence's completion to become visible; release the page. The hardware obligation is that any device transaction whose translation is decided after the completion becomes visible must not use a translation the invalidation removed.

Note what is not required. Before the invalidation is issued, the IOMMU is entitled to answer from its cache with the old translation: the cache is architecturally permitted to be stale until told otherwise. A test suite that treats staleness itself as a bug tests the wrong thing; the property is about the boundary, not about caching.

Note also what makes this hard to test after the fact. The failure is a translation that should not have been produced, and translations are not observable from software: the device sees data, not addresses, and by the time the wrong data is noticed the page has been reused and the evidence is gone. A stale-translation bug in silicon presents as memory corruption in an unrelated subsystem, under load, without a reproducer. This is an argument for testing the property where translations are observable, which is at RTL, on the IOMMU's completion port.

C. Why the window is not narrow

It is tempting to assume the exposure is a few cycles wide and therefore unlikely to matter. It is not. The device is not paused while software invalidates; a DMA engine with outstanding requests keeps issuing transactions across the whole sequence, and a driver that unmaps a buffer typically does so while other buffers belonging to the same device are still active. The exposure lasts as long as a stale entry survives in the cache, which, if the invalidation missed it, is until something else evicts it. This is why our strongest test keeps traffic in flight rather than quiescing the device, and it is why the distinction between P5 and P5′ matters: they can be satisfied and violated independently.

III. Related Work

Litmus testing is the established method for probing memory ordering. Suites and generators exist for RISC-V's memory model [8], and tools run them on hardware and against axiomatic models. RTLCheck [3] connects such models to RTL by translating litmus outcomes into assertions verified against a processor design, and RealityCheck [4] adds modularity to that flow. Recent work scales hardware consistency verification further [5].

Translation-aware ordering is the subject of COATCheck [1], which models ordering at the hardware–OS interface including TLB and page-table events, and TransForm [2], which synthesises litmus tests for those "transistency" models. These address a CPU's translation hardware and the operating system that maintains it.

A. The software side of the same boundary

Operating systems have long treated IOMMU invalidation as expensive, and the cost has driven design choices that make the hardware property more rather than less load-bearing. The IOTLB invalidation bottleneck and strategies for reducing it are well documented [7], and the strategies all amount to invalidating less often: batching invalidations, deferring them until a queue fills or a timer expires, and reusing addresses in ways that assume the invalidation eventually lands. Linux exposes this directly as a choice between a strict mode, which invalidates and waits on every unmap, and a deferred mode, which does not.

Both modes depend on the property we test, and the deferred mode depends on it more sharply. In strict mode the wait follows every unmap, so the window is short and frequent. In deferred mode the system accumulates a batch of unmapped addresses, flushes them together, and then releases all of those pages at once; a single missed entry in that flush releases a page that the device can still reach, and the release happens in bulk, at a moment chosen by the allocator rather than by the driver. The performance work that motivates deferred invalidation is sound, and it makes the hardware's ordering guarantee the thing holding the whole scheme up.

B. The gap

What is missing is the device side. An IOMMU's translation caches are filled by the IOMMU itself, invalidated through a command queue rather than by instructions, and observed only indirectly through the addresses the device's transactions acquire. None of the above tests that path against hardware, and and the verification efforts that do exist for these devices [9], [10] report coverage of functional translation rather than ordering under invalidation.

IV. Ordering Properties

We state six properties, P1 through P5 together with P5′, a variant of P5 that must hold under concurrent traffic. Each is decided by observing the address that leaves

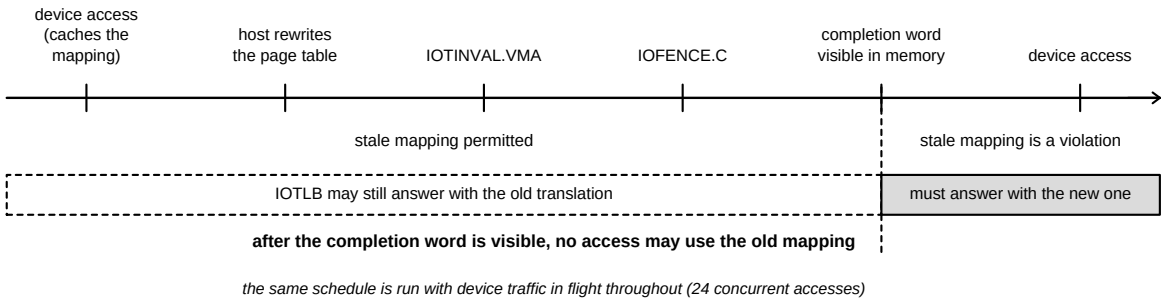


Fig. 1. The schedule underlying P5. Before the fence completion is visible the IOMMU may answer from its cache; after it, it may not. The same schedule is run with device traffic in flight throughout.

the IOMMU's completion port for a device transaction, so each is testable on any implementation without internal visibility.

P1 (staleness permitted). If a page-table entry is modified and no invalidation is issued, a transaction to that page may be answered with the old translation. This is the control: it confirms that the cache is active and that the later properties are not vacuously satisfied by an implementation that simply never caches.

P2 (effectiveness). After an unscoped invalidation, a transaction to the affected page must be answered with the new translation, and the IOMMU must have re-read the page tables to obtain it.

P3 (scope precision). After an invalidation scoped to one address, a transaction to that address must be answered with the new translation.

P4 (scope containment). The same scoped invalidation must not disturb other cached translations: a transaction to a different cached page must still be answered from the cache. P3 and P4 together pin the scope from both sides; an implementation that ignores scope and flushes everything satisfies P3 and fails P4, and one that ignores the command satisfies P4 and fails P3.

P5 (completion ordering). Once the fence's completion word is visible in memory, no subsequent transaction may be answered with a translation removed by a command preceding the fence (Fig. 1).

P5′ (completion ordering under load). P5 must hold when device transactions are in flight across the whole sequence, not only when the device is quiescent while software invalidates.

V. Method

We realise the properties against the open RISC-V IOMMU IP published by Zero-Day Labs [11], using a Verilator environment that drives the device's four AXI interfaces, serves its implicit page-table and command-queue accesses from a memory image, and observes translated addresses on the completion port. The environment and its construction are described separately [12]; what matters here is that the testbench owns the memory image, so it can modify a page-table entry between two device transactions and can poll for the fence completion word directly.

Each property becomes a schedule. For P1: map a page, access it so the translation is cached, rewrite the leaf entry to a different physical page, access it again, and require the old address. For P2: repeat, but enqueue an unscoped IOTINVAL.VMA first, and require both the new address and evidence that a walk occurred. For P3 and P4: cache two pages, rewrite both entries, invalidate only the first, and require the first to be refreshed and the second to be unchanged. For P5: rewrite an entry, enqueue the invalidation followed by IOFENCE.C with a completion address, poll the image for the completion word, and only then issue the transaction.

Three details make these schedules trustworthy, and each was a correction to an earlier version of the environment.

First, the page-table rewrite must not be visible to the IOMMU as anything but a memory change. The testbench writes the new leaf entry directly into its memory image rather than through any hardware path, so a cached translation can only be refreshed by the IOMMU re-reading the entry.

Second, "the translation was refreshed" must be distinguished from "the translation happened to be right." The environment counts the implicit accesses the IOMMU issues on its data-structure port, so P2 can require both the new address on the completion port and evidence that a walk occurred; an implementation that returned the correct address without re-reading memory would be caching something it should not.

Third, the fence completion must be observed the way software observes it. The testbench polls its memory image for the completion word at the address supplied in the command, exactly as a driver polls the completion location, rather than watching an internal signal. This matters because the property is about what is visible to software, and an internal "invalidation done" signal may assert earlier than the word appears.

P5′ uses two concurrent processes. One issues a stream of twenty-four transactions to the affected page with

TABLE I
Ordering properties on the implementation under test

	Property	Result
P1	staleness permitted before invalidation	holds
P2	invalidation is effective (forces re-walk)	holds
P3	scoped invalidation refreshes the named page	holds
P4	scoped invalidation leaves other pages cached	holds
P5	no stale translation after fence completion	holds
P5′	P5 under twenty-four concurrent transactions	holds

TABLE II
Injected invalidation defects

Defect	Checks failed	Caught by
IOTLB entry not cleared on a matching invalidation	6	P2
Unscoped invalidation clears only the first IOTLB entry	4	P2*

*only by the multi-page form of P2.

a short gap between them. The other waits, enqueues the invalidation and fence, and sets a flag when the completion word appears. Every transaction the first process completes after that flag is set is checked against the new translation; transactions completed before it are unconstrained. The schedule is arranged so that a useful number of transactions fall on each side of the boundary—if all of them landed before the fence the test would be vacuous—and the test records the split, so a run in which too few transactions were post-fence is visible rather than silently weak.

VI. Results

Table I summarises. All six hold. Thirteen of the fifty-six checks in the wider suite belong to this family.

Quantitatively: under P2 the invalidated access re-reads the page tables, which the environment confirms by counting the implicit accesses the IOMMU issues. Under P4 the untouched page is still answered from the cache with its original translation, so the scope is neither too wide nor ignored. Under a variant of P2 that caches four pages—filling the IOTLB in the configuration under test—and then invalidates all of them, all four are refreshed. Under P5 the fence completion word appears twelve cycles after the command is enqueued, and the subsequent transaction uses the new translation. Under P5′, twenty of the twenty-four transactions complete after the fence completion becomes visible, and none of them uses the stale translation (Fig. 2, upper row).

The last figure is worth reading as a statement about the property rather than about the design. To the left of the dashed line the two rows are indistinguishable, and correctly so: nothing is being claimed there. Everything the property says is to the right of it, and that is exactly where the two rows separate completely.

VII. Do the Tests Discriminate?

A suite in which everything passes is worth little unless it can be shown to fail when it should. We injected two invalidation defects into the RTL, each a single scripted edit, rebuilt and reran.

The first makes invalidation ineffective: the IOTLB entry is not cleared when a matching invalidation arrives. Six checks fail, P2 among them.

The second is subtler and is the reason P2 was extended to the multi-page case. It makes an unscoped invalidation clear only the first IOTLB entry, leaving the rest stale—the shape of a real partial-flush bug. It is detected by four checks.

It is worth being precise about why, because our first account of this was wrong. We assumed a single-page test would miss the defect, since the one page it checked would be the entry that was cleared. It does not: the single-page form of P5 catches it too, because the page that test uses happens not to occupy the surviving entry. That is luck, not design—move the test's address so that it lands in entry zero and the defect goes unseen. What the multi-page form provides is not detection the single-page form lacks, but detection that does not depend on which cache entry an address happens to land in.

Table II records both. Under the partial-flush defect, P5′ reports nineteen stale translations out of nineteen post-fence transactions (Fig. 2, lower row): once the surviving entry is the one the device is using, every transaction after the fence is wrong. That is what the failure looks like in hardware, and it is the reason we consider the concurrent variant worth the extra machinery—it is the test that turns a single wrong address into an unmistakable pattern.

We draw two methodological points from this. The first is that the multi-page form of P2 is not redundancy; it is the only thing separating a suite that catches partial-flush defects from one that does not, and single-page invalidation tests are the natural thing to write. The second is that both defects are single-line edits that leave the design fully functional—it translates, it walks, it faults correctly—which is a fair model of how such a bug reaches silicon. Neither is caught by any test outside this property family.

VIII. The Guarantee Is Incidental

The implementation satisfies P5, but the reason deserves scrutiny. Its command handler implements IOFENCE.C by writing the completion word when the command is processed; it does not wait for preceding commands to retire. The source states the reasoning directly, and anticipates the problem:

> "I think this command makes sense when implementing ATS commands, or any other command that could take several cycles to execute. […] Since all implemented commands in this version are executed immediately, there's no need to wait for now."

The comment is right on both counts. The fence is for long-latency commands, and there are none yet.

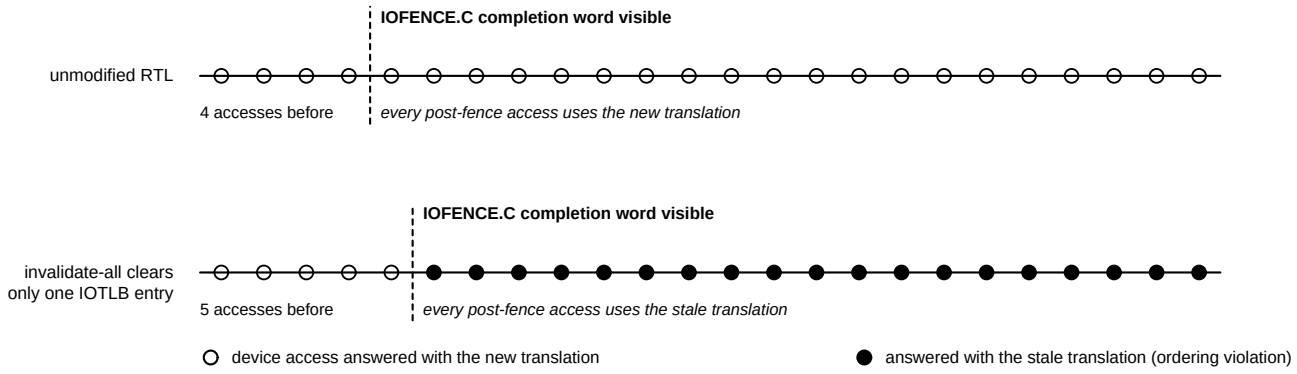


Fig. 2. P5′ on the unmodified design and on a design with a partial-flush defect injected. Each marker is one device transaction. The property constrains only the transactions to the right of the dashed line. The defect is invisible before the fence and total after it.

That argument is sound for the current feature set and unsound as a design property. IOFENCE.C exists precisely to order commands whose effects are not immediate. The specification's ATS invalidation commands, which this implementation does not provide, require a round trip to a device and are exactly the long-latency case the fence is for; the same applies to any future command whose effect is pipelined rather than combinational. When such a command is added, P5 will no longer follow from the structure of the design, and nothing in the current implementation will notice: the fence will keep completing immediately and the tests will keep passing until a command is slow enough for the window to open. The implementation also marks the fence's predecessor-read and predecessor-write bits as unimplemented.

This is the practical value we would claim for the property family. P5 as tested is satisfied today. Kept in a regression suite, it becomes a guard on the assumption that makes it true, and it will fail on the day that assumption stops holding.

IX. Applying the Properties Elsewhere

Nothing in P1–P5′ is specific to RISC-V. Each is stated in terms of three capabilities that any IOMMU offers in some form: a way to cause a translation to be cached, a way to invalidate it, and a way to learn that the invalidation has taken effect. An ARM SMMU expresses the second and third as TLBI commands followed by a synchronisation command whose completion is polled in a consumer register; an Intel VT-d unit expresses them as invalidation descriptors followed by an invalidation-wait descriptor that writes a status word to memory, which is close enough to IOFENCE.C that the P5 schedule transfers almost unchanged. The scoping distinctions behind P3 and P4 map onto those architectures' domain- and address-selective invalidations.

What must be rebuilt for each target is the observation path: the properties are decided by watching the physical address an IOMMU emits, so they need an environment that can see that port and that owns the memory image the IOMMU walks. That is the bulk of the work, and it is work a verification team doing anything else with the design has already done. Added to an existing environment, the property family is a few hundred lines.

Two things would have to be reconsidered for a design with a deeper pipeline than the one we tested. P5′ assumes that a transaction observed at the completion port after the fence word appeared was in fact decided after it; in a design that speculates on translations or that reorders completions, that inference needs an additional argument or an internal probe. And a design with multiple independent translation caches—per-device or per-context—needs P4 restated over each of them, since an invalidation that reaches one cache and not another is precisely the defect the property is meant to catch.

X. Limitations

This is a study of one implementation, and a negative result in the sense that no violation was found. The properties are stated operationally rather than formally; we do not claim they are complete, and a formal treatment in the style of the transistency work would likely yield more. Only first-stage invalidation is exercised: second-stage (IOTINVAL.GVMA) and directory-cache (IODIR) invalidation are not, nor is invalidation scoped by address-space or virtual-machine identifier, nor are ATS invalidations, which the implementation does not provide. Concurrency is exercised with a single device stream; multiple devices with distinct identifiers invalidating and translating simultaneously would be a stronger test. All results come from simulation of one implementation with one simulator, at one configuration of IOTLB and directory-cache sizes; the multi-page form of P2 uses four pages because that fills the cache in this configuration, and the number would have to be re-chosen for a larger one.

Finally, fault injection measures whether the tests catch the defects we thought to inject, which is evidence that they discriminate but not a bound on what they would miss.

XI. Conclusion

The ordering obligation an IOMMU owes to an operating system's unmap path is testable, cheaply, against RTL, and it is not covered by the CPU-side work on translation-aware memory ordering. Six properties, expressed as schedules over the command interface and decided by watching the completion port, are enough to pin the boundary from both sides—requiring staleness to end after the fence, and requiring scoped invalidation not to over-reach. An open RISC-V IOMMU satisfies all of them, including under concurrent traffic, and two injected invalidation defects are caught. The implementation's fence, however, does not wait for preceding commands; its compliance rests on every implemented command being immediate, which is a fact about today's feature set rather than a guarantee of the design.

The result we would most like to be acted on is the cheapest one. These are six tests. They run in seconds, they need no formal model and no proprietary tools, and they belong in the regression suite of any IOMMU under development—not because they are likely to fail today, but because the assumption that makes them pass is the kind that is quietly invalidated by a feature added two years later by someone who never read the comment explaining why the fence did not need to wait.

Artifact Availability

The tests, the environment they run in, and the fault-injection harness are released under a permissive licence and run with open tools.

References

[1] D. Lustig, G. Sethi, M. Martonosi, and A. Bhattacharjee, "COATCheck: Verifying memory ordering at the hardware–OS interface," in ASPLOS, 2016.

[2] N. Hossain, C. Trippel, and M. Martonosi, "TransForm: Formally specifying transistency models and synthesizing enhanced litmus tests," in ISCA, 2020.

[3] Y. A. Manerkar, D. Lustig, M. Martonosi, and M. Pellauer, "RTLCheck: Verifying the memory consistency of RTL designs," in MICRO, 2017.

[4] Y. A. Manerkar, D. Lustig, and M. Martonosi, "RealityCheck: Bringing modularity, hierarchy and abstraction to automated microarchitectural memory consistency verification," arXiv:2003.04892, 2020.

[5] G. Ravi, X. Qiu, M. Thottethodi, and T. N. Vijaykumar, "QED: Scalable verification of hardware memory consistency," arXiv:2404.03113, 2024.

[6] A. T. Markettos, C. Rothwell, B. F. Gutstein, A. Pearce, P. G. Neumann, S. W. Moore, and R. N. M. Watson, "Thunderclap: Exploring vulnerabilities in operating system IOMMU protection via DMA from untrustworthy peripherals," in NDSS, 2019.

[7] N. Amit, M. Ben-Yehuda, and B.-A. Yassour, "IOMMU: Strategies for mitigating the IOTLB bottleneck," in Workshop on Interaction between Operating Systems and Computer Architecture (WIOSCA), ISCA, 2010; published in LNCS 6161, Springer, 2012.

[8] "litmus-tests-riscv: RISC-V architecture concurrency model litmus tests." [Online]. Available: https://github.com/litmus-tests/litmus-tests-riscv

[9] M. Rodriguez (Zero-Day Labs) and S. Waheed (10xEngineers), "Open-source commercial-grade RISC-V IOMMU with verification," RISC-V Summit North America, 2024.

[10] A. A. Petrykin, I. A. Stotland, and A. N. Meshkov, "Standalone verification of IOMMU with virtualization supporting," Proc. Institute for System Programming of the RAS, vol. 31, no. 3, pp. 77–84, 2019.

[11] Zero-Day Labs, "RISC-V IOMMU IP," commit 18ad30a. [Online]. Available: https://github.com/zero-day-labs/riscv-iommu

[12] M. Timbadiya, "Verifying the RISC-V IOMMU: an open verification environment and a fault-injection evaluation," companion article.